

ALGEBRAIC MODELS FOR SPACES

Meeting Timings: Wednesdays 13h-15h

Meeting Venue: 7.07, Hans Freudenthalgebouw

Organisational Meeting: 9th September

RATIONAL HOMOTOPY THEORY

Talk 1. Overview of rational homotopy theory (23.09; Vignesh). Give a survey of rational homotopy theory: explain rationalisation of nilpotent spaces and prove Quillen's theorem $\mathrm{coCAlg}(Ch_{\mathbb{Q}})^{\geq 2} \simeq S_{\mathbb{Q}}^{\geq 2} \simeq \mathrm{Lie}(Ch_{\mathbb{Q}})^{\geq 1}$. Survey Sullivan's polynomial de Rham complex functor A_{PL} , minimal models and the statement that rational cochains is fully faithful for finite type spaces. If time remains, do some example computations with minimal models.

Quillen's original reference is [Qui69] and Sullivan's [Sul77]. The following surveys can be helpful: [FH17], [Hes07] and [Heu19, Section 2] to connect to the point of view in talk 7. A slightly different approach is explained in [Lur11].

Talk 2. Rational dichotomy (30.09; Laurent). Explain and prove the dichotomy theorem: rational homotopy groups of finite spaces are either of finite total dimension as a graded vector space (then the space is *elliptic*) or they grow exponentially (the space is *hyperbolic*); see [FHT01], but also shorter surveys like [Hes07, Section 2.4] and [Ber12] or [FHT82, Thm. 1.1]. Discuss examples of elliptic and hyperbolic spaces and their properties. For example, elliptic spaces have non-negative Euler characteristic and satisfy Poincaré duality; see [Fél89, Section 5]. If time remains, present some of the consequences for rational Lie algebras [FHT82, Thm. 1.2 for example].

p -ADIC HOMOTOPY THEORY

We address the following question: what other subcategories/localisation of the category of spaces admit algebraic descriptions? First, we turn towards p -complete spaces and then discuss some integral solutions.

Talk 3. Mandell's theorem and the coalgebraic model (7.10; Karolis). Briefly introduce p -complete spaces and prove Mandell's theorem [Man01] stating that the functor $C^*(-, \bar{\mathbb{F}}_p) : (S^{\leq 2, \mathrm{fin}})^{\mathrm{op}} \rightarrow \mathrm{Alg}_{E_{\infty}}(Ch_{\bar{\mathbb{F}}_p})$ is fully faithful and describe its essential image. Discuss [BB26], explain how passing to coalgebras allows to remove the finiteness assumption and what the main added difficulties are. Mention why there is no hope for a full p -adic Lie algebra picture; see the introduction of [BB26]. If time remains, explain that the integral cochain functor is faithful but not full [Man06]. We will see different fixes to this in the following talks.

Talk 4. Integral models via the higher Frobenius (14.10; Ryan). Survey [Yua23]. In particular, explain and prove theorems B and C, giving an integral cochain model for simply connected finite spaces via *Frobenius fixed E_{∞} -rings*. Explain why assembling the rational and $\bar{\mathbb{F}}_p$ -cochains is not feasible and how the

p -adic model from theorem B fixes this problem. Highlight the role played by the Tate-valued Frobenius.

Talk 5. Binomial rings and integral homotopy types (21.10; Guy). This talk should discuss [Hor24] and [Ant23].

The aim is to present a different approach to fix the issue raised by Mandell that integral cochains are not full by considering *binomial structures*. Consider $C^*(-, \mathbb{Z})$ as a functor to commutative cosimplicial rings rather than to E_∞ -algebras. Horel observes that $\mathbb{Z}^X$ is degreewise a *binomial ring* and that integral cochains enhanced with their binomial structure give a fully faithful model for finite-type nilpotent spaces.

Parallely, survey the results in [Ant23]. Here, instead of animating binomial rings, one considers the ∞ -category of derived δ -rings and trivialises their Frobenius to also obtain an integral model for spaces; see [Ant23, Thm. 1.4, Thm. 1.3].

Talk 6. Schematization (28.10; Francesca). Explain Grothendieck’s schematisation of homotopy types problem as described in *Pursuing Stacks* and in [Toe06, Appendix A]. Then discuss [Toë20], in particular the description of the homotopy groups of “affine integral homotopy types”. If time remains, mention the discussion around the Grothendieck-Teichmüller group in [Hor24]. Another direction could be to highlight some results from the rational setting from this perspective as done in [Lur11], in particular mentioning coaffine stacks.

PERIODIC HOMOTOPY THEORY

Talk 7. Lie algebra model for v_n -periodic spaces (4.11; Rens). Recall that in talk 1 we had seen that rational spaces are equivalent to connected rational Lie algebras. Rational spaces are v_0 -periodic spaces and it turns out that more generally v_n -periodic spaces are equivalent to $T(n)$ -local Lie algebras. This talk’s aim is to present that result following [Heu21]. More precisely, introduce v_n -periodic spaces, discuss the Bousfield-Kuhn functor and prove [Heu21, Thm. 2.6].

Talk 8. The Behrens-Rezk model and comparison map (11.11; Maite). Discuss and prove [BR17, Thm. 1.1], the Behrens-Rezk comparison map; see [BR20] for the more technical paper. Explain the notions appearing in the theorem, e.g. TAQ, introduce $\Phi_{K(n)}$ -good spaces, give examples and counter-examples; see [BH19]. The talk should focus on Sections 5, 6 and 7 of [BR17].

OUTLOOK

Talk 9. Open Questions (18.11; Gijs). Explain some further directions and applications that these algebraic models for spaces could have. What are some interesting open questions in this directions?

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